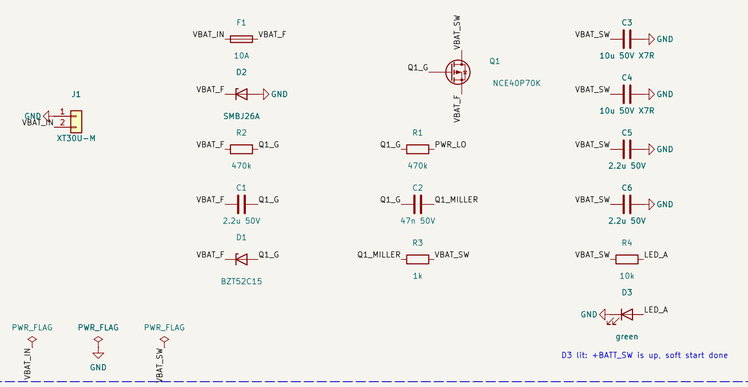
Battery in, main fuse, soft start

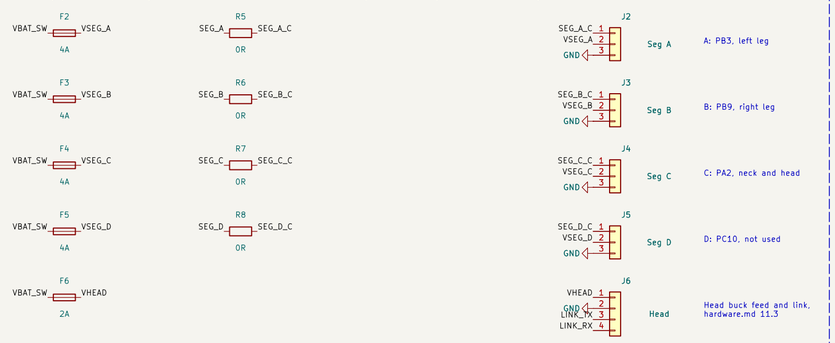
XT30U-M hand fitted. Q1 ramps +BATT_SW over about 30 ms (C2 Miller ramp); C1 holds Q1 off at plug-in. docs/bridge-pcb.md 3.3



D3 lit: +BATT_SW is up, soft start done

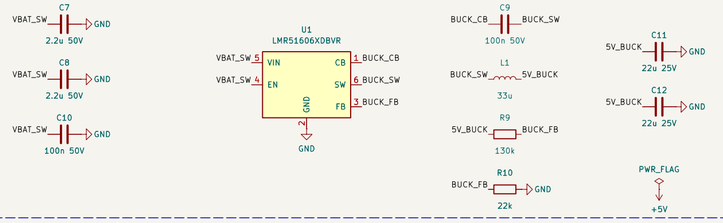
Branches: servo segments A to D, head

Fuses time lag (BHFUSE BSMD1206C). Data lines: 0R links, room for a series resistor (hardware.md 5.6). PH 2.0 vertical through hole, hand fitted



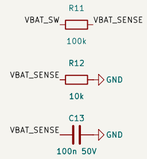
5 V buck for the WeAct board

LMR51606X, 400 kHz. Values from TI Table 8-1: 130k/22k gives 5.53 V, about 5.2 V after D8. EN tied to VIN



Pack voltage to PA0

Full scale 36 V (hardware.md 5.8)



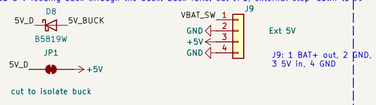
Mounting holes

M2, one per corner, not connected

H1 M2 H2 M2 H3 M2 H4 M2

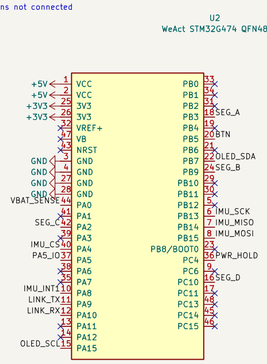
Buck output, backup 5 V

D8 stops USB 5 V feeding back through the buck. Buck fails: cut JP1, external step-down to J9



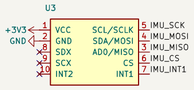
Bridge MCU: WeAct STM32G474 core board

Soldered flush. Pin allocation: hardware.md 5.8. Unused pins not connected



Trunk IMU: LSM6DSV16X breakout

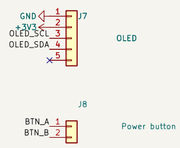
SPI2, 4-wire. Soldered flush, 7-pin row only



LSM6DSV16X breakout

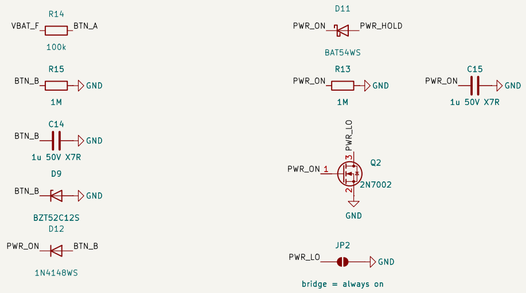
Display and button connectors

PH 2.0 vertical through hole, hand fitted



Soft power switch (docs/bridge-pcb.md 6.5)

Button or PC4 turns Q2 on, which grounds R1 and lets the soft start turn Q1 on. Q2 off: Q5 turns Q4 on, snapping Q1 off. JP2 bridged = always on, as before. Off leaves about 20 uA of leakage

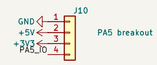


Q3: PB5 reads low while the button is pressed

J8 (button) joins BTN_A and BTN_B

PA5 breakout

Wire pads, bottom-left corner



Bootstrapped by scripts/kicad/gen_bridge_sch.py: edit this file from now on
Design note: docs/bridge-pcb.md

Sheet: /
File: bridge.kicad_sch

Title: Dukki bridge carrier

Size: A2	Date: 10/10/2026	**Rev: A, draft**
KiCad E.D.A. 10.0.6		Id: 1/1